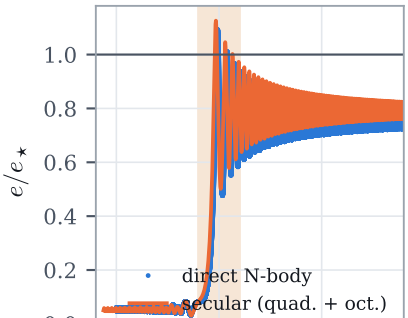
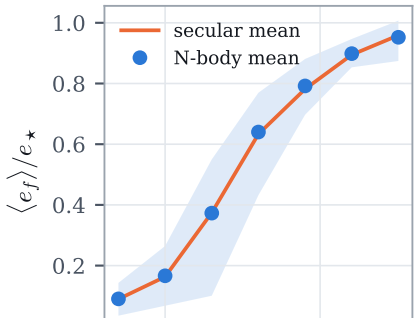
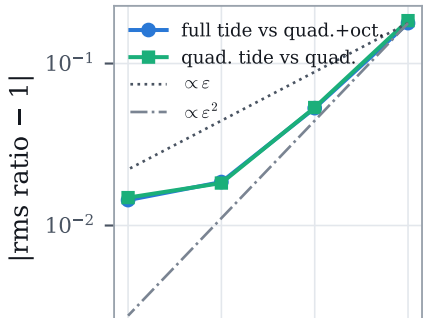

(a) one trajectory, $\varepsilon = 0.005$, $\Lambda = 8$
$e/e_\star$
δ/c
direct N-body
secular (quad. + oct.)


(b) ensembles, $e_0 = 0.05\, e_\star$
secular mean
N-body mean
$\langle e_f \rangle / e_\star$
Λ


(c) convergence, $\Lambda = 1$
full tide vs quad.+oct.
quad. tide vs quad.
$\propto \varepsilon$
$\propto \varepsilon^2$
|rms ratio − 1|
$\varepsilon = n_p/n_m$